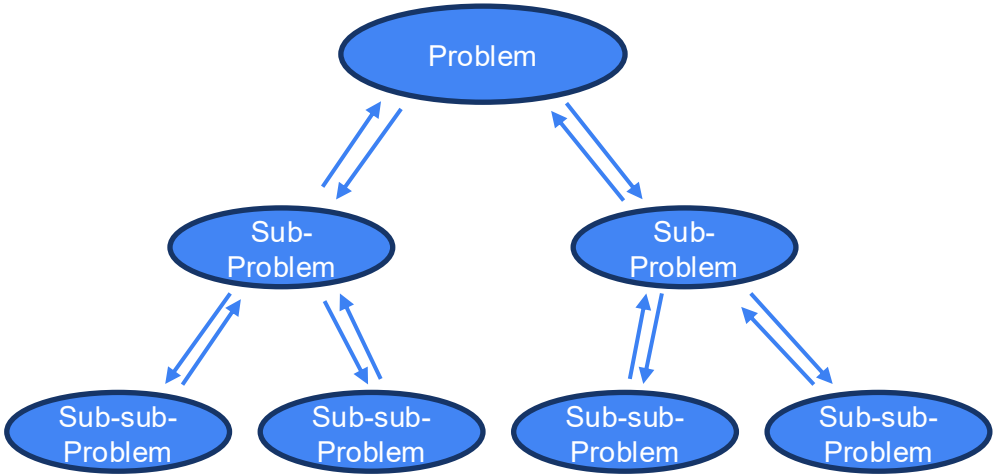
Lecture 14

Divide and Conquer I



Announcements

- Homework 5 due this Sunday night (3/29)
- Group Meetings continue
  - New groups next week
- Individual Project 2 due next Sunday night (4/5)

Greedy Algorithms

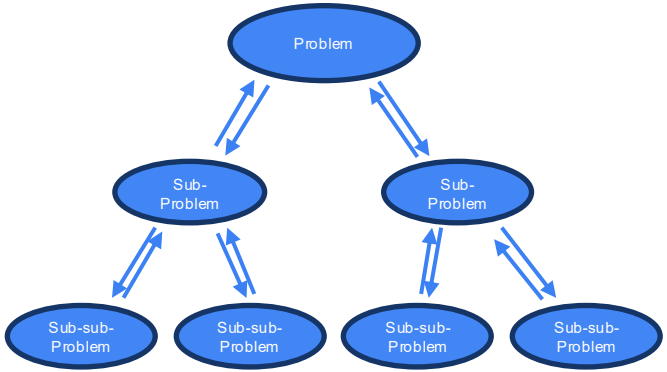
We are moving on to our study of algorithm design

- Greedy
- **Divide-and-conquer**
- Dynamic Programming
- Network Flow

Divide and Conquer Idea

General Recipe (or pattern)

- Divide problem into several parts
- Solve each part recursively
- Combine solutions to sub-problem into overall solution



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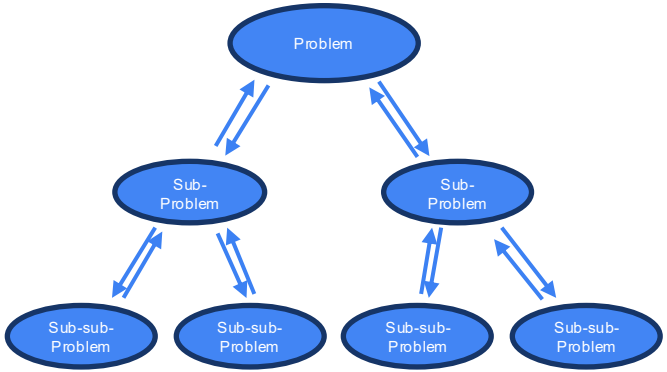
Our Focus

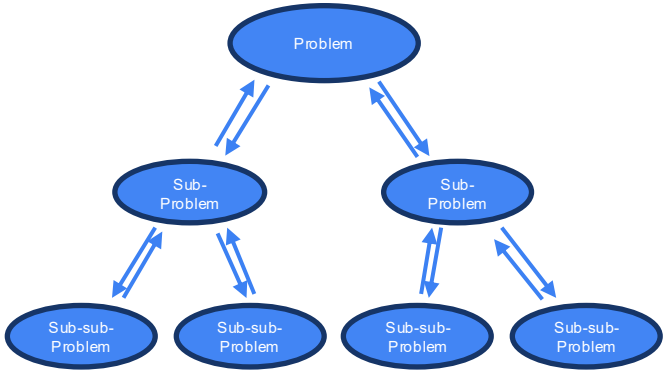
- Algorithm design
- Running time analysis



Divide and Conquer Idea

Typical Setting

- Divide problem of size n into two subproblems of size $n/2$
- Solve two subproblem recursively
- Combine two solutions into overall solution



Running Time Analysis

- Brute force: $\Theta(n^2)$
- Divide-and-conquer: $O(n \log n)$

Merge Sort

Problem: given a list L of n unique items, arranger them in ascending order

$\mathrm{MergeSort}(L)$:

if L has one item **do**

return L

Divide L into two halves A, B

$A \leftarrow \mathrm{MergeSort}(\mathrm{A})$

$B \leftarrow \mathrm{MergeSort}(\mathrm{B})$

$L \leftarrow \mathrm{Merge}(A, B)$

return L

Merge Sort

Problem: given a list L of n unique items, arranger them in ascending order

Let $T(n)$ be the worst-case running time of $\mathrm{MergeSort(L)}$

$\mathrm{MergeSort}(L)$:

if L has one item **do**

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Base case: $O(1)$

Divide L into two halves A, B

$A \leftarrow \mathrm{MergeSort(A)}$

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Recursive cases: $T(n/2)$

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Combination step: $\Theta(n)$

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Recursive cases: $T(n/2)$

Combination step: $\Theta(n)$

$$T(n) = 2T\left({}^{n}/_{2}\right) + \Theta(n)$$

Recurrence

$T(n) = 2T\left({}^{n}/_{2}\right) + O(n)$

- $T(n)$ is defined in terms of small values of n
- For base case: $T(1)$ is $O(1)$

Goal: "solve" the recurrence by finding a simple expression for $T(n)$ for all n

Recurrence

Recurrence:

- $T(n) = 2T(^{n}/_{2}) + O(n)$
- $T(1) = O(1)$

Let's the use definition on Big-O:

- $T(n) \leq 2T(^{n}/_{2}) + cn$
- $T(1) \leq c$

Exercise 1

Recurrence:

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Q: Why can we use the same value of c in both instances of the Big-O definition?

a) Take $c = \min(c_1, c_2)$ where c_1 and c_2 are the values from each instance

b) Take $c = \max(c_1, c_2)$ where c_1 and c_2 are the values from each instance

c) It's not okay!

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Strategies for Solving:

1. Unrolling recursion
2. Recursion tree (more structured unrolling)

Unrolling

Idea: "unroll" the recurrence

$$
\begin{aligned}
T(n) &\leq 2T(n/2) + cn \\
&\leq 2[2T(n/4) + c(n/2)] + cn \\
&= 4T(n/4) + 2cn \\
&\leq 4[2T(n/8) + c(n/4)] + 2cn \\
&= 8T(n/4) + 3cn \\
&\leq \cdots \\
&\ldots \\
&\leq nT(1) + \log_2 n \cdot cn = O(n \log n)
\end{aligned}
$$

Exercise 2

Q: suppose we have the recurrence $T(n) = T(n/2) + T(n/3)$. What do we get after two unrollings?

a. $T(n/4) + T(n/9)$

b. $T(n/4) + 2T(n/6) + T(n/9)$

c. $2T(n/6)$

d. $T(n/4) + T(n/6) + T(n/9)$

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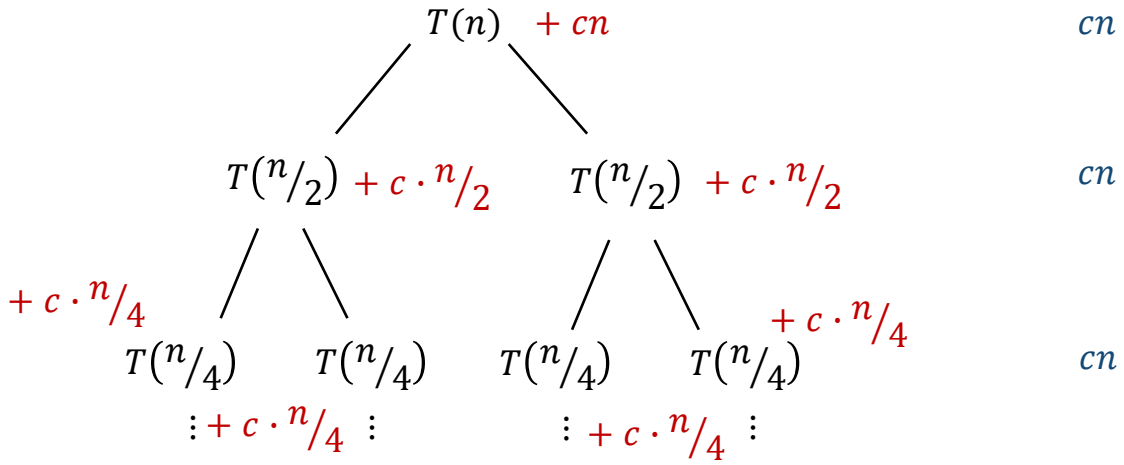
Recursion Tree



$\log_2 n + 1$ levels each with cn work

- $T(n) \leq cn(\log n + 1)$ implies $T(n)$ is $O(n \log n)$

Guess and Verify

Another approach is to guess the running time and proof it using induction

- We're going to skip this one but can talk about it during office hours

Maximum Subsequence Sum

Input: a list L of n numbers

Goal: find the value of the largest subsequence sum (including empty subsequence)

Example

- $L = [\underbrace{4, -1, 8, -2, 3}, -5, 1, -5]$

- MSS: 14 (sum of first five elements)

Exercise 3

Which of the following are true for a maximum sum subsequence?

a. It has more positive than negative numbers

b. It does not start or end with a negative number

c. Any maximal subsequence of negative numbers is bordered by a subsequence of positive numbers with a larger sum in absolute value

Exercise 3

Which of the following are true for a maximum sum subsequence?

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Baseline MSS Algorithm

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- Idea: check all subsequences

```
MSS_baseline(L):
    n ← len(L)
    Initialize A as an n × n matrix of zeroes
    for i = 1 to n do
        sum ← 0
        for j = i to n do
            sum += L[j]
            A[i, j] ← sum
    return max_{i,j} A[i, j]
```

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Running time: $O(n^2)$

Can we do better?

Divide-and-Conquer for MSS

Recursive algorithm for MSS

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Recursive algorithm for MSS

Idea:

- Find MSS X in left half of list
- Find MSS Y in right half of list
- Find MSS Z for subsequences that cross the midpoint of list

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$$X = 13;\ Y = 3; Z = 14$$

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- Return maximum of X, Y, Z
- How do we find X, Y, Z?

Divide-and-Conquer for MSS

$\text{MSS_dc}(L, low, high)$:

if $high == low$ **do**

return $L[low]$

$mid \leftarrow \left\lfloor \frac{low+high}{2} \right\rfloor$

$X = \text{MSS_dc}(L, low, mid)$

$Y = \text{MSS_dc}(L, mid + 1, high)$

return $\max(X, Y, Z)$

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$sum \leftarrow 0; X' \leftarrow 0$

for $i = mid$ to low **do**

$sum \mathrel{+}= L[i]$

$X' = \max(sum, X')$

$sum \leftarrow 0; Y' \leftarrow 0$

for $j = mid + 1$ to $high$ **do**

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$Y' = \max(sum, Y')$

$Z = X' + Y'$

return $\max(X, Y, Z)$

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Recursive cases

$sum \leftarrow 0; X' \leftarrow 0$

for $i = mid$ to low **do**

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$X' = \max(sum, X')$

Compute left part of Z

$sum \leftarrow 0; Y' \leftarrow 0$

for $j = mid + 1$ to $high$ **do**

$sum \mathrel{+}= L[j]$

$Y' = \max(sum, Y')$

Compute right part of Z

$Z = X' + Y'$

Compute Z

return $\max(X, Y, Z)$

Return max

Running Time

Let $T(n)$ be the running time of $\mathrm{MSS_dc}(L)$ on a list of size n

- Base case is $O(1)$
- Two recursive calls on arrays of size $n/2$: $2T(n/2)$
- Work outside of recursive calls: $O(n)$

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Implies the running time is $O(n \log n)$

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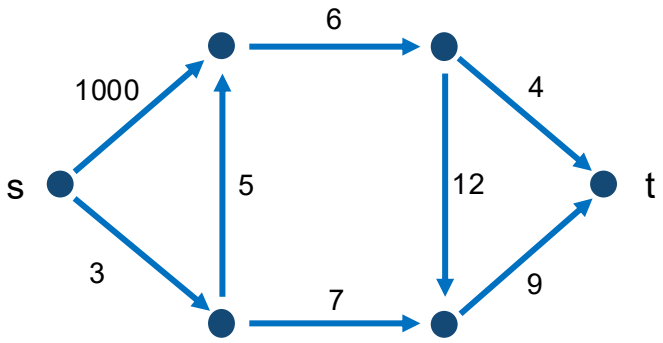
Shortest path problem

- Find shortest paths in a directed graph with edge weights (lengths)
- **Problem**: can we efficiently find $d(v)$ for all vertices $v \in V$?



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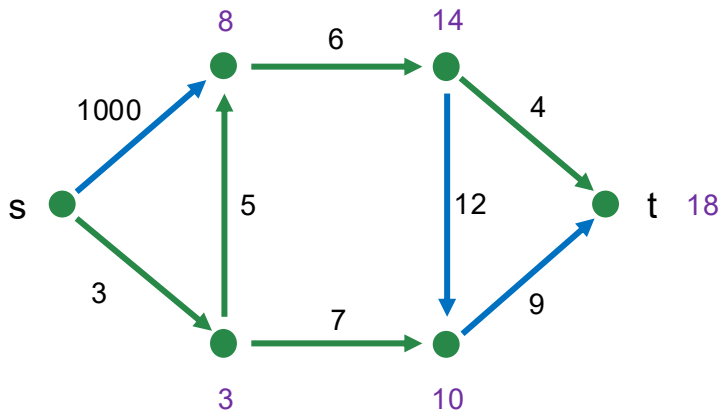

Dijkstra's Algorithm

- Idea: explore the "nearest" vertex from s
- Weighted version of BFS
- The shortest path from s to t has length 18
- Running time: $O\big((m+n)\log n\big)$

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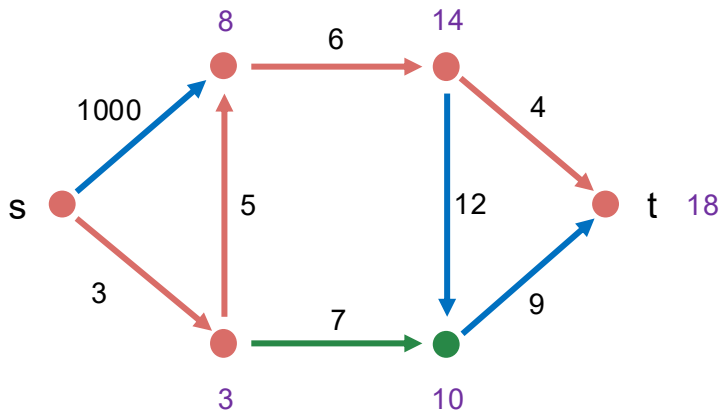

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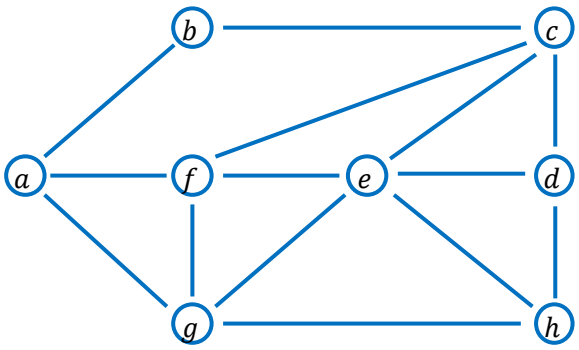
Cycles and Cuts

Recall some graph properties:

- $abcdef$ is a **path** from a to f
- $abcdefa$ is a **cycle**



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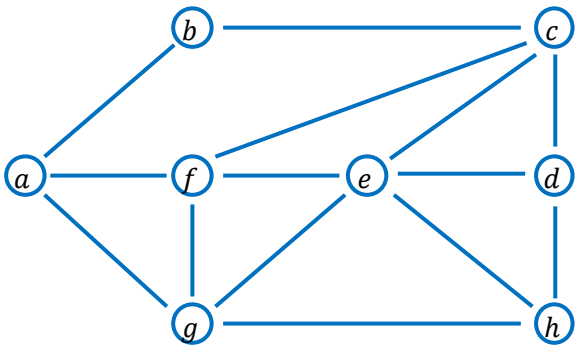
Here are some new properties:

- A **cut** is a partition of the vertices into two non-empty subsets S and $V \backslash S$



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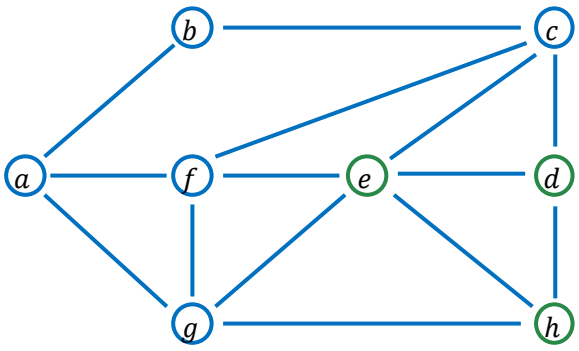

$S = \{e, d, h\}$

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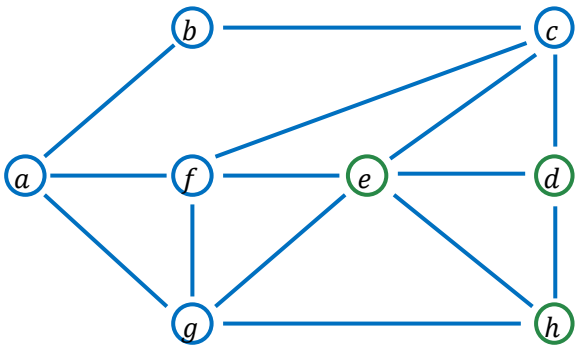

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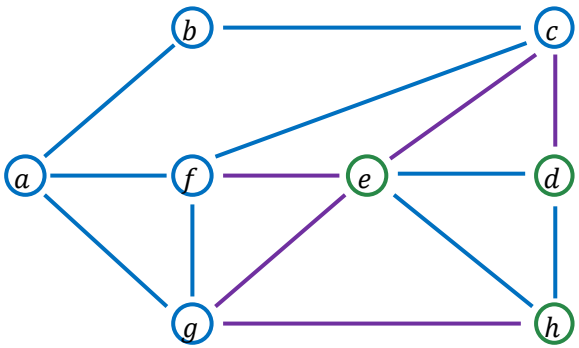
- Deleting a cutset D of edges disconnects the subsets of a cut



$$S = \{e, d, h\}$$

$$D = \{(e, g), (e, f), (c, e), (c, d), (g, h)\}$$

Exercise I

Consider the cut $S = \{b, f, g, d\}$. Which edge is in the cutset of S?

a. (b, f)

b. (g, f)

c. (f, e)

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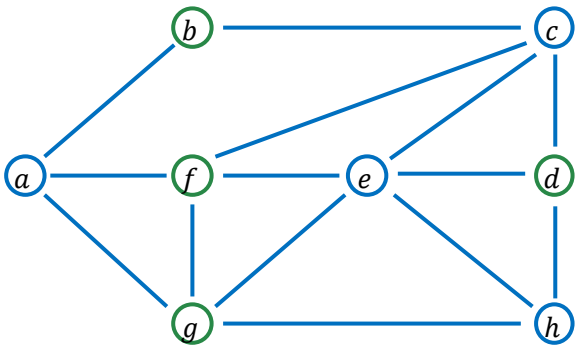

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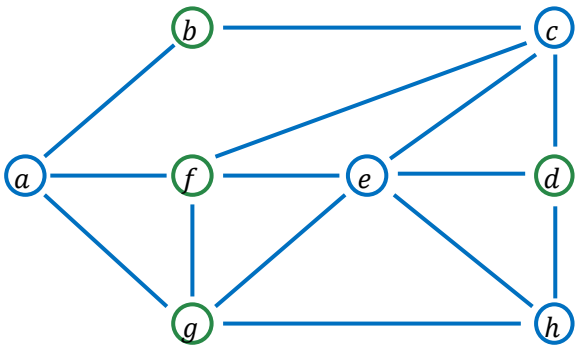
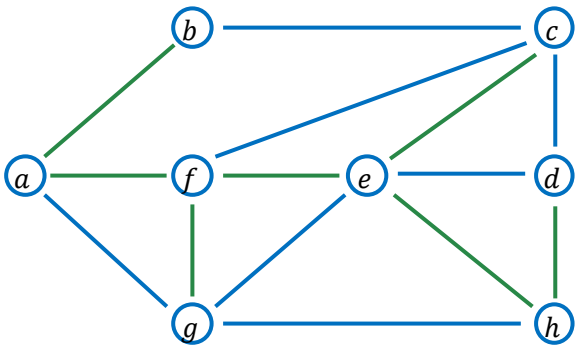


Spanning Trees

Let $H = (V, T)$ be a subgraph of an undirected graph $G = (V, E)$. Then H is a **spanning tree** of G if H is both acyclic and connected.

- In other words: a spanning tree is a subgraph on all the vertices that is a tree
- Where have we seen spanning trees before?



Exercise II

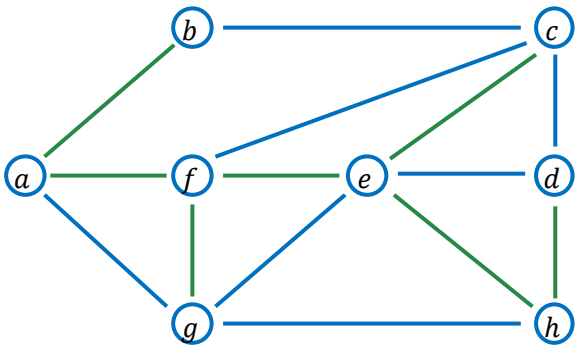
Which of the following properties are true for all spanning trees H?

a. H contain exactly $|V| - 1$ edges

b. Removing any edge from H disconnects it

c. Adding any edge to H creates a cycle

d. All of the above



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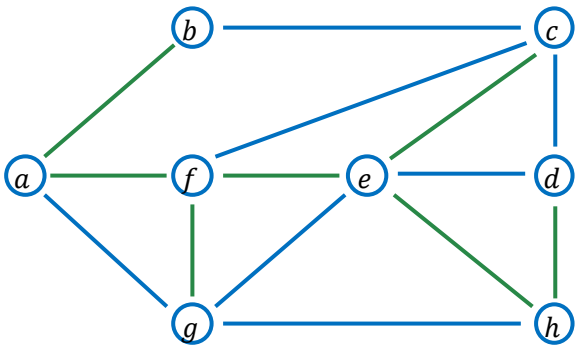
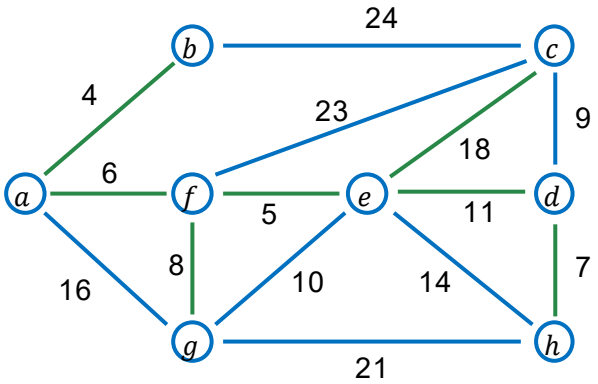


Minimum spanning tree (MST)

Given a connected, undirected graph $G = (V, E)$ with edge weights c_e, a **minimum spanning tree** (V, T) is a spanning tree of G such that the sum of the edge costs in T is minimized

- In other words: an MST is a spanning tree with the lowest edge costs



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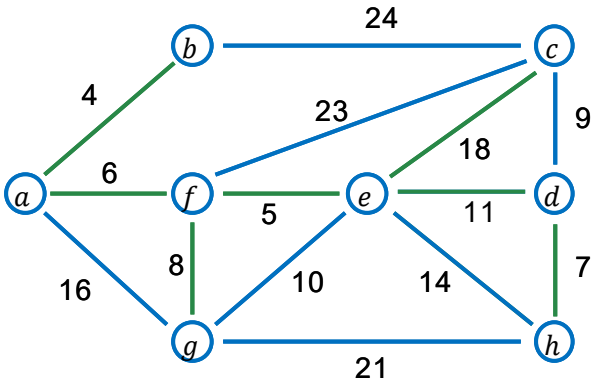
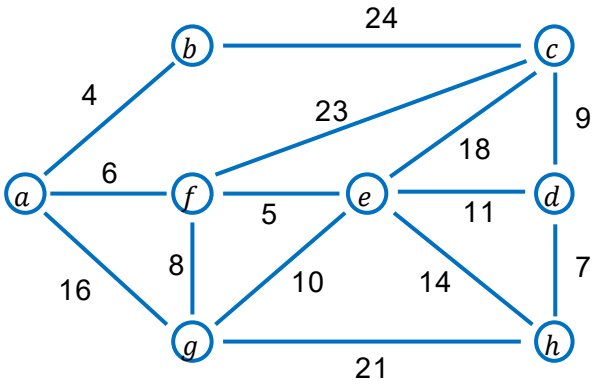
- In other words: an MST is a spanning tree with the lowest edge costs
- Finding an MST is a fundamental problem in CS with diverse applications



Cut Property

Th: Let D be the cutset of cut S in graph G. Then, the minimum weight edge e in D belongs to every MST of G.

- e is the *cheapest* or *lightest* edge in the cut
- It is *safe* to add e to an MST



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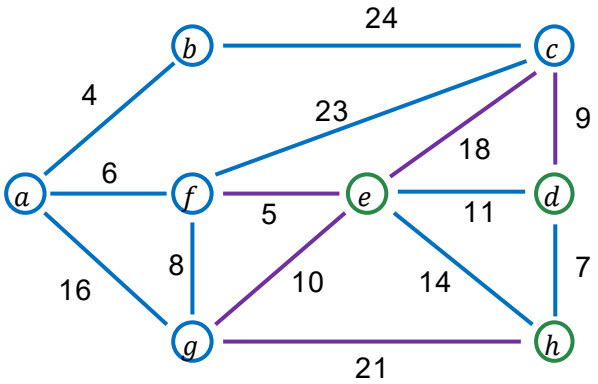

$$S = \{e, d, h\}$$

$$D = \{(e, g), (e, f), (c, e), (c, d), (g, h)\}$$

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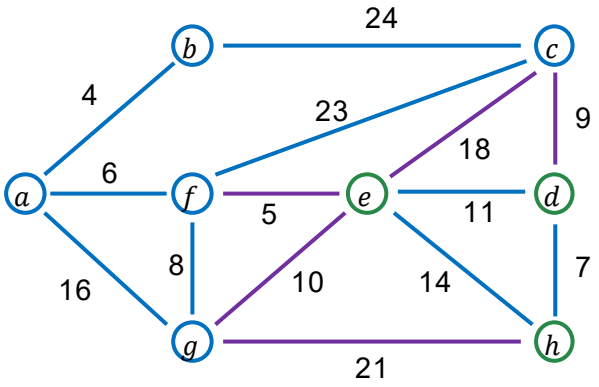
- Prove using contradiction by supposing e is the minimum weight edge in the cutset for cut S but e is not in an MST.
- Good idea for Project II



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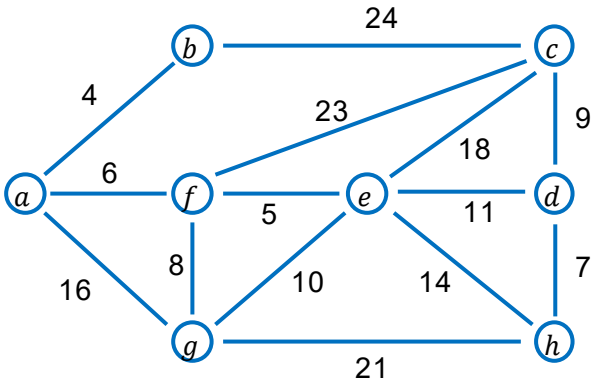
Kruskal's Algorithm

Idea: add edges in order of weight so long as it does not create a cycle



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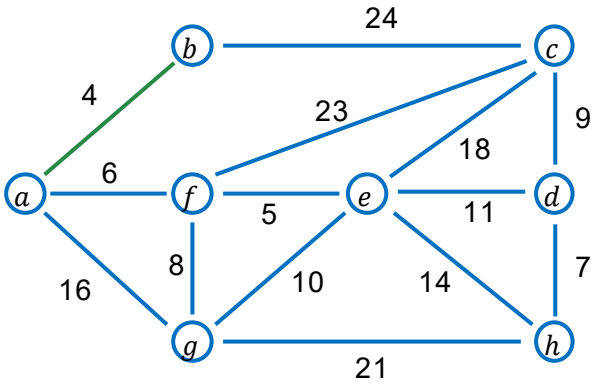

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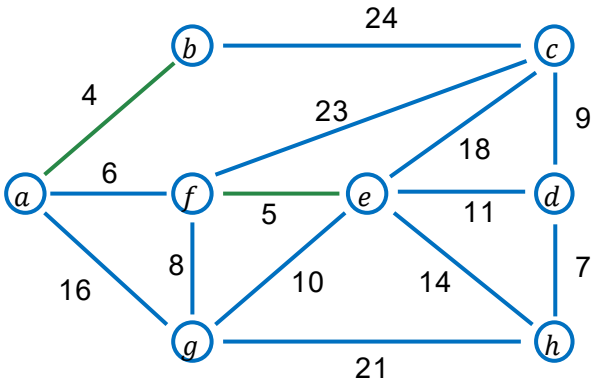

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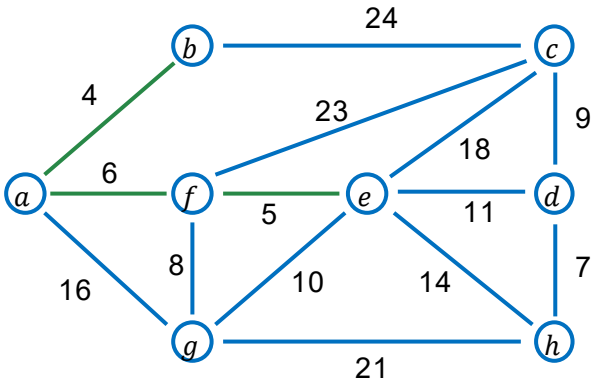

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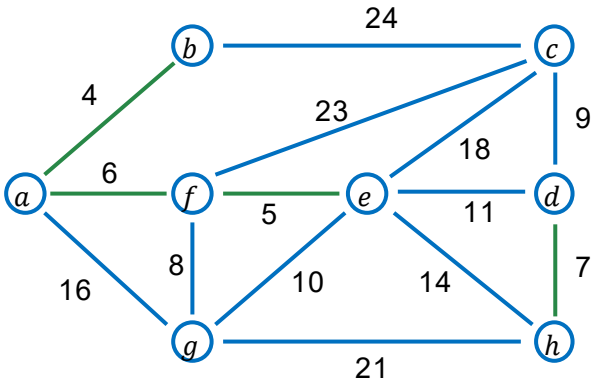

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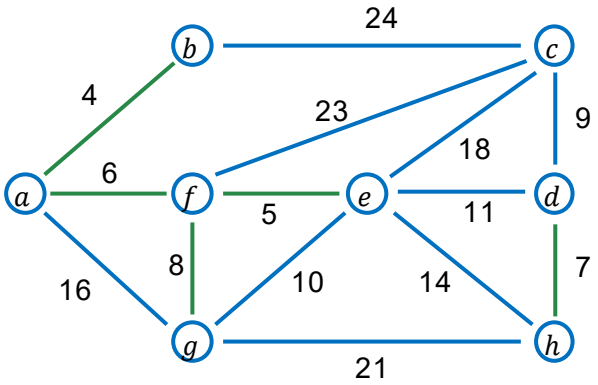
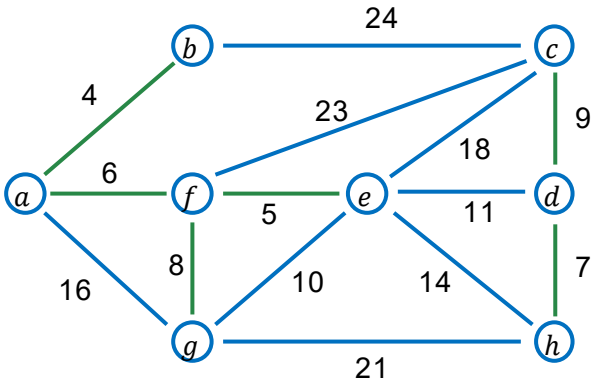


Kruskal's Algorithm

Idea: add edges in order of weight so long as it does not create a cycle

How does this use the cut property?



Kruskal's Algorithm

Idea: add edges in order of weight so long as it does not create a cycle

Assume edges are numbered $e = 1, \dots, m$

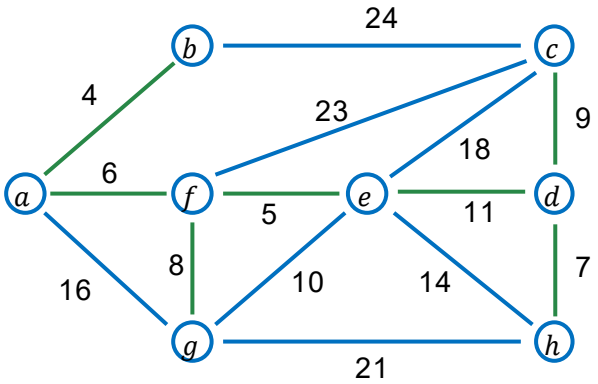
Sort edges by weight so $c_1 \leq \cdots \leq c_m$

Initialize $T = \{\quad\}$

for $e = 1$ to m **do**

if adding e to T does not form a cycle **then**

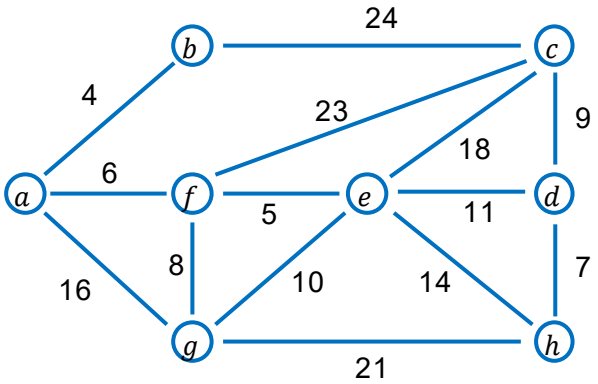
$T = T \cup \{e\}$



Prim's Algorithm

Idea: grow a tree as a single connected component from a vertex s

- Let S be the component containing s
- Add the cheapest edge from S to $V \backslash S$



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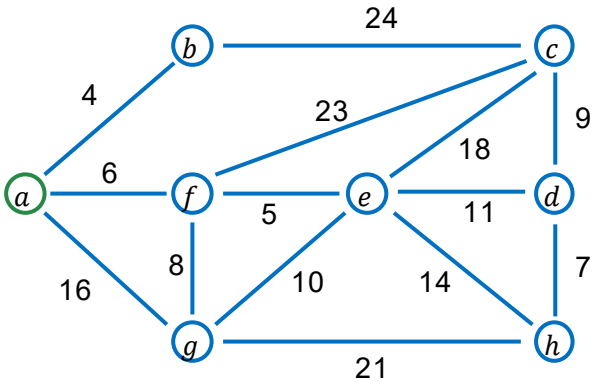

$S = \{a\}$

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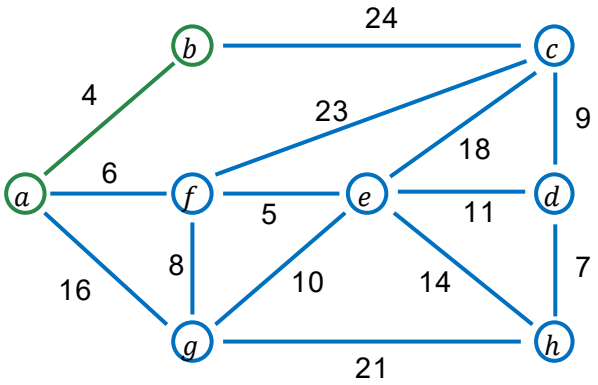

$S = \{a, b\}$

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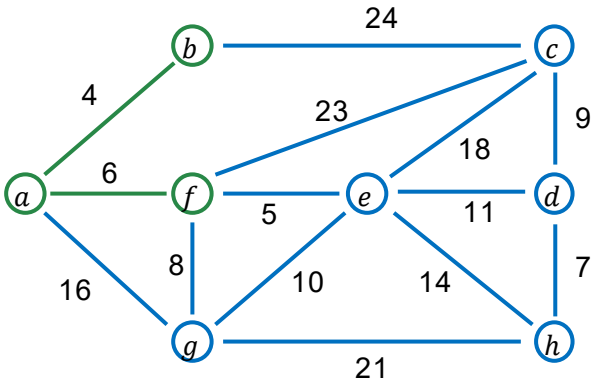

$S = \{a, b, f\}$

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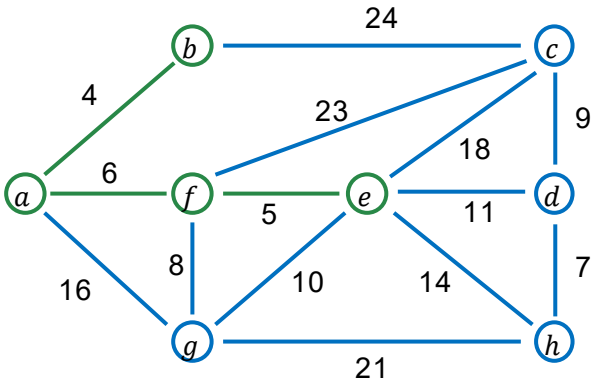

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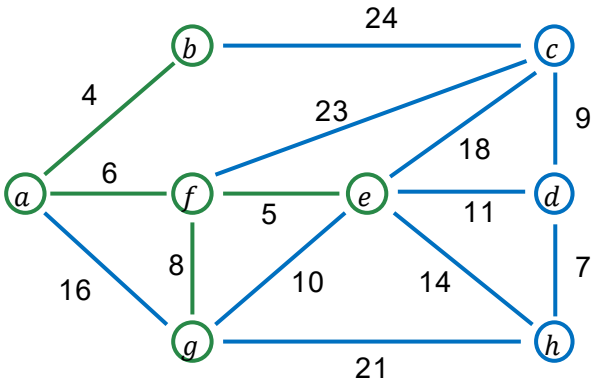

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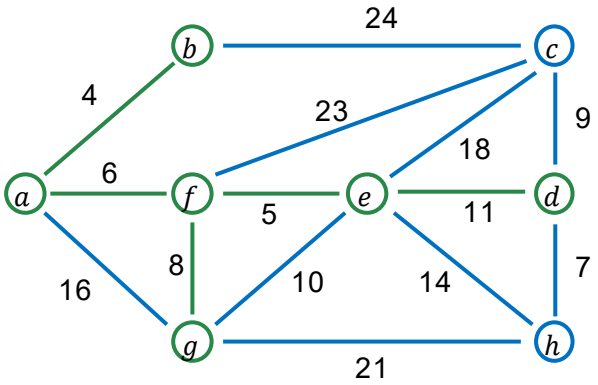

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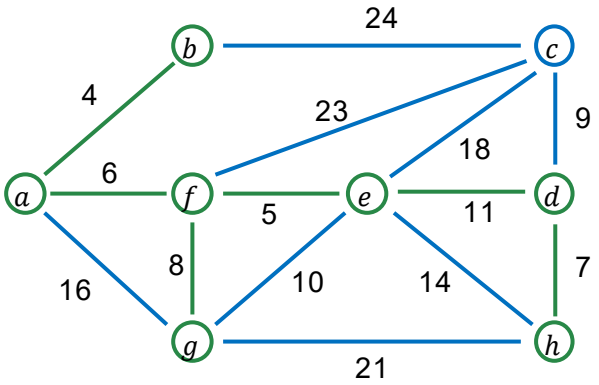

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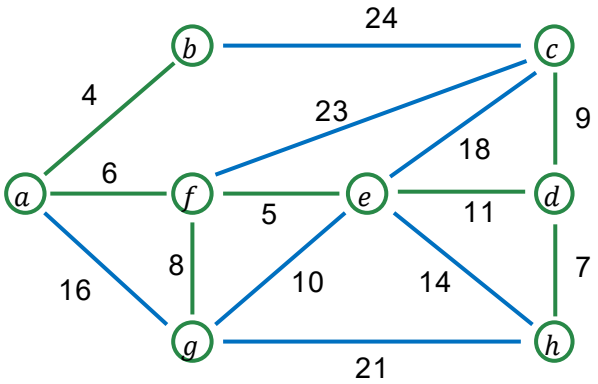

$S = \{a, b, f, e, g, d, h, c\}$

Next Time

- Dynamic Programming